

Supplementary Materials

Can Grip-Force Dynamics Track Perceptual Strain? A Trial-Level Spectral Analysis in Older Adults

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Main manuscript: [manuscript.qmd](#) (HTML/PDF) · [manuscript-biorxiv.qmd](#) (bioRxiv preprint PDF)

1 Signal processing and feature extraction

Grip-force preprocessing followed the pipeline documented in the main Methods and in `analysis/01_ingest.qmd`:

1. Target-normalized error: $e(t) = (F(t) - F_{\text{target}})/F_{\text{target}}$
2. Linear detrending within each trial epoch
3. Fourth-order zero-phase Butterworth high-pass filter at 0.5 Hz (`sosfiltfilt`)
4. Welch PSD estimation (Hann window, 50% overlap, ~0.5 Hz resolution)
5. Band integration: visuomotor tracking (0.5–3.0 Hz) and physiological tremor (8.0–12.0 Hz)

Epochs spanned grip-cue onset through 1 s after probe onset. All spectral features were log-transformed before modeling.

2 Model specifications

Mixed-effects logistic regression. Perceptual accuracy was modeled as:

$$\text{logit } P(\text{correct}) = \beta_0 + \beta_1 \text{modality} + \sum_{k=1}^4 \beta_k \text{poly}_k(\text{difficulty}) + \beta_5 z(\text{tracking}) + \beta_6 z(\text{tremor}) + u_i$$

where $u_i \sim \mathcal{N}(0, \sigma^2)$ is a random intercept for participant i , and $z(\cdot)$ denotes within-condition standardized band power. Primary inference used high-grip trials only (40% MVC); low-grip models used the same structure.

Cross-validated machine learning. Random-forest and logistic classifiers were trained with subject-wise cross-validation (leave-one-subject-out) on high-grip trials. Feature sets compared:

- **Baseline:** stimulus difficulty level, task modality
- **DSP:** baseline features plus tracking- and tremor-band power

Performance was evaluated using area under the ROC curve (AUC).

Table 1: Cross-validated AUC for perceptual accuracy prediction (high-grip trials).

Feature set	Model	AUC	N trials
Baseline	Logistic	0.6170	7001
Baseline	RandomForest	0.8498	7001
DSP	Logistic	0.6279	6911
DSP	RandomForest	0.8553	6911

3 Sample coverage and attrition

Table 2: Grip-force data attrition funnel from behavioral oddball tasks to analysis-ready trials.

Stage	Subject- tasks	Tri- als	% retain- able	Note
Oddball subject-tasks in behavioral v3	136	18,099	—	aud + vis; runs merged into trial counts
Excluded: aborted mid-task	13	478	—	beh_trials < 100
Retainable subject-tasks	123	17,621	100.0%	individual missed trials already removed in v3
No grip files (entire subject $\times$ task)	1	128	0.7%	zero grip CSVs found; all behavioral trials in block excluded
Partial grip (behavioral > on-disk grip)	36	3,678	20.9%	some grip files exist but fewer trials than behavioral (often 1 run = 30 trials)
Grip trials on disk	122	14,430	81.9%	fatigue-break runs merged per session; retainable subject-tasks only
Feature extraction loss	95	671	3.8%	grip trials present but spectral features not extracted (short epochs, etc.)
Analysis set (trial features extracted)	122	13,759	78.1%	primary grip force modeling cohort

4 Additional figures and tables

Single-trial signal chain. Representative high-grip incorrect trial showing the transformation from raw force to normalized error, rolling variance, and Welch PSD with analysis bands marked.

Probe-locked spectrogram. Time–frequency power aligned to probe onset ($t = 0$), with normalized error in the upper panel.

Population spectral distributions.

Manipulation checks and individual differences.

Tremor-band power and perceptual accuracy.

Predictive modeling and brain–behavior.

Full fixed-effects table.

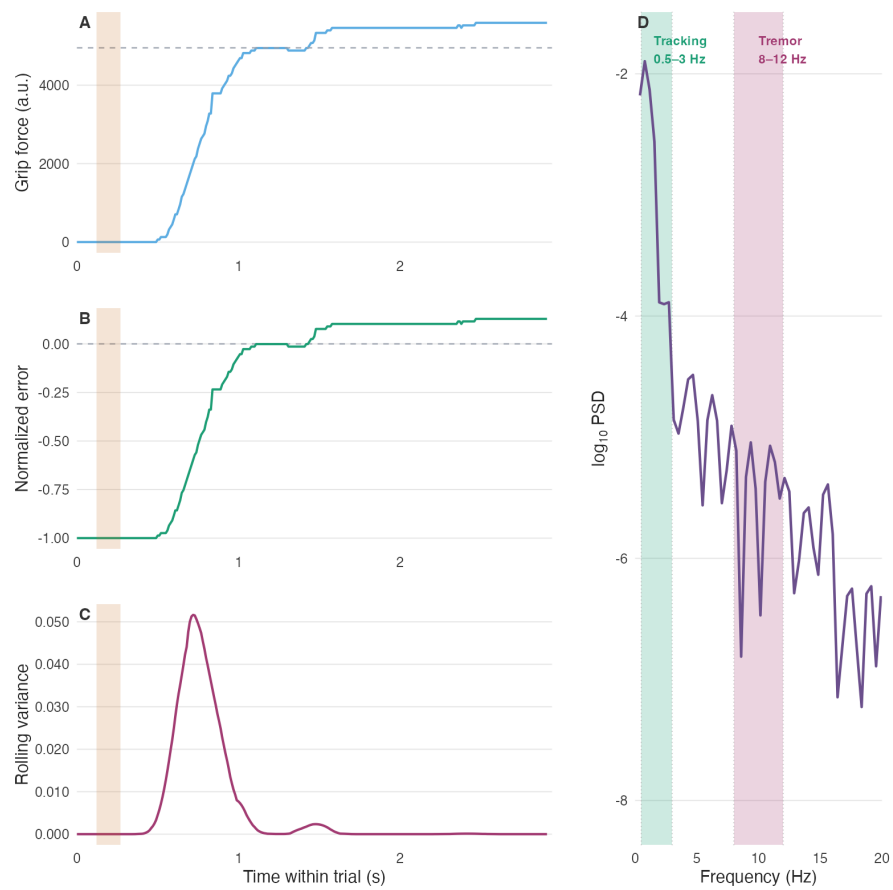


Figure 1: Single-trial signal chain from grip force to spectral phenotype.

Representative Probe-Locked Force Trace and Short-Time Spectrum

BAP171 · vis · session 2 · trial 8 · incorrect response

A = normalized force error; **B** = short-time spectrum (representative trial; not inferential). Vertical gold line = oddball probe onset ($t = 0$). Green band = tr

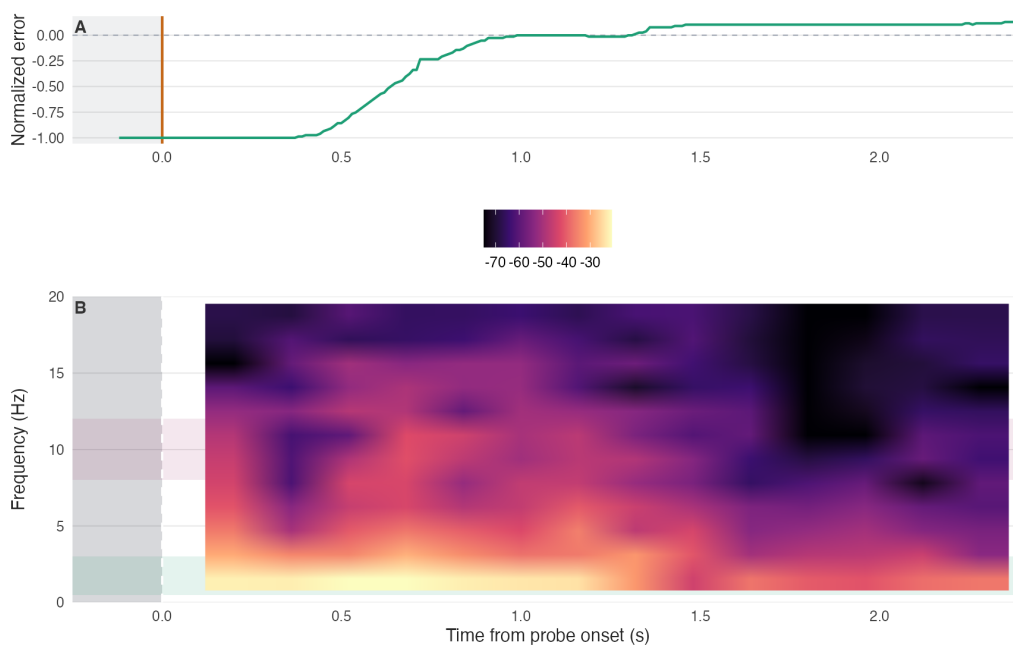


Figure 2: Probe-locked motor spectrogram aligned to oddball onset.

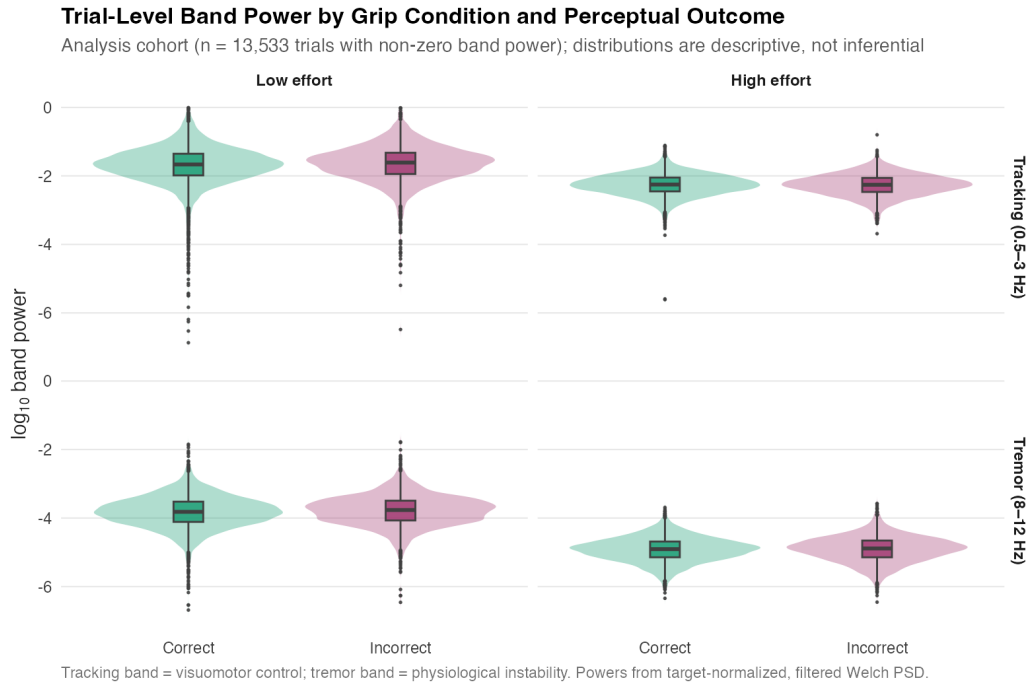


Figure 3: Trial-level log band-power by grip effort and perceptual outcome.

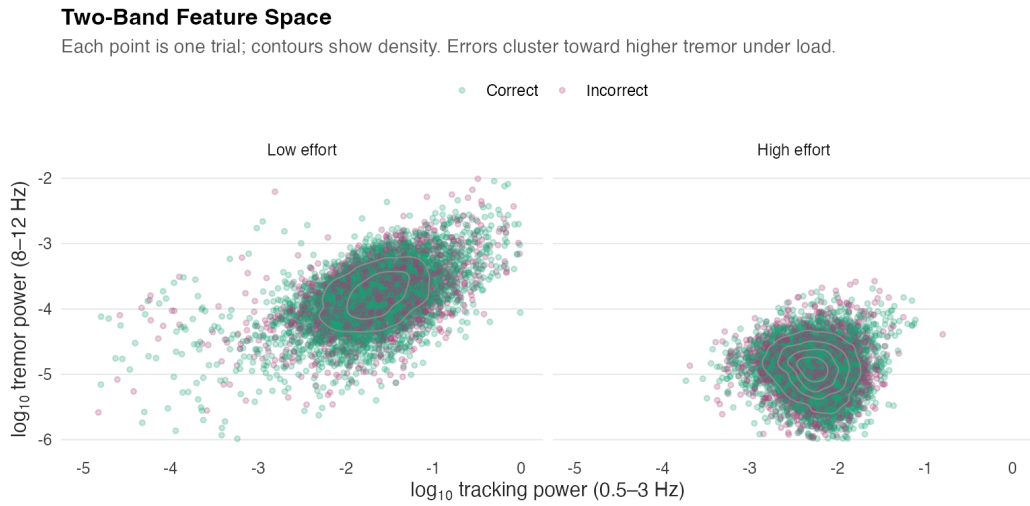


Figure 4: Two-band feature space (tracking vs. tremor), split by grip effort.

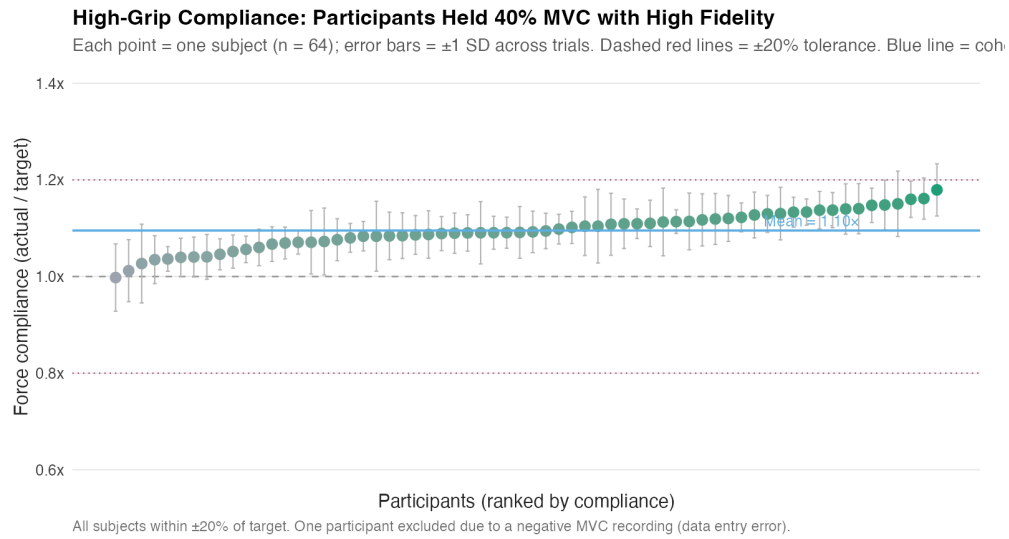


Figure 5: Per-subject high-grip force compliance (actual / target). All participants maintained force within $\pm 20\%$ of the 40% MVC target. One participant with a negative MVC recording was excluded from all spectral analyses.

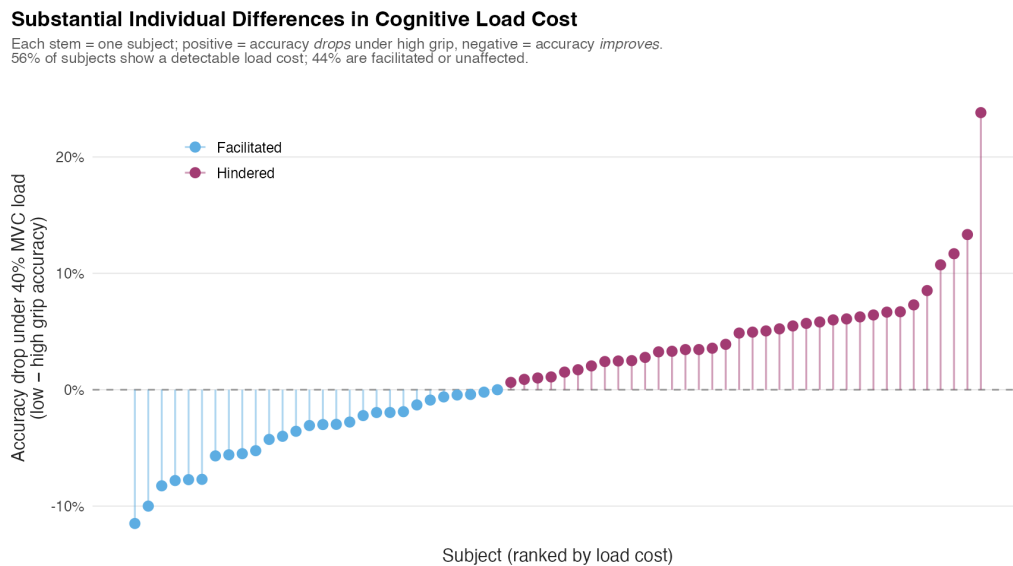


Figure 6: Subject-level accuracy drop under high versus low grip. High grip produced lower accuracy for the majority of participants.

Load-Related Accuracy Differences Were Not Explained by Faster Responses

Each line connects one subject's SAT point under low (light blue) vs. high (magenta) grip. Diamond markers = grand mean $\pm$ SE. High grip shifts the cloud *down* (lower accuracy) with little systematic change in RT, arguing against a pure speed-accuracy trade-off.

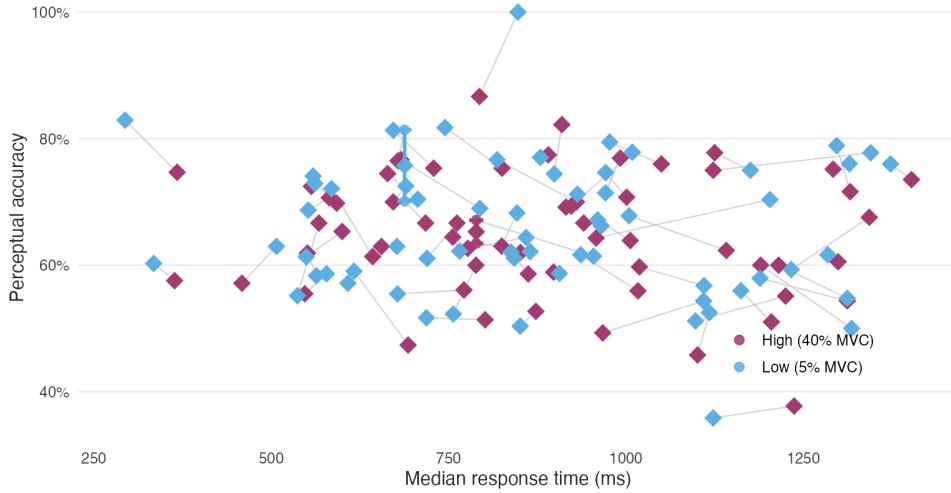


Figure 7: Speed-accuracy trade-off under low versus high grip. Load-related accuracy differences were not accompanied by systematically faster responses.

Tremor-Band Power and Perceptual Accuracy by Grip Effort

Marginal accuracy from mixed-effects model (modality + difficulty adjusted; population-level predictions)

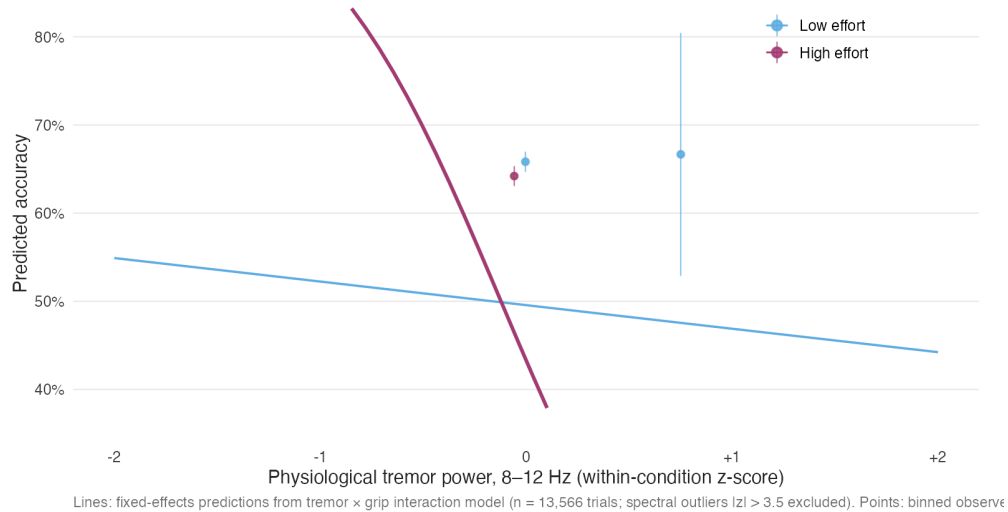


Figure 8: Marginal perceptual accuracy by tremor-band power quantile, split by grip effort. Lines show model-predicted values; points show binned observed accuracy $\pm$ 95% CI. Both conditions show a negative association. Shown for visualization; the tremor $\times$ grip interaction was not significant ($p = .244$).

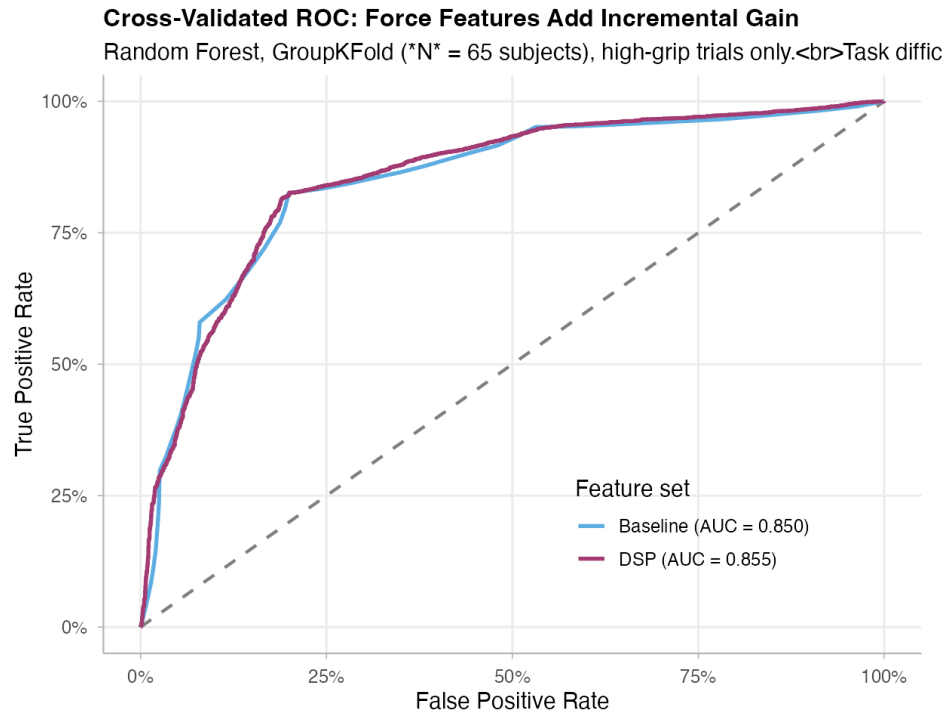


Figure 9: Cross-validated ROC curves (random forest, high-grip trials).

LC Microstructure and Tremor Reactivity to Physical Load

Subjects with lower LC fICVF show larger tremor increases under high grip ($r = -0.21$, $N = 54$). CNR shows a near-zero relationship ($r = -0.04$), consistent with prior evidence that diffusion metrics better capture functional microstructure than neuromelanin contrast in non-clinical samples.

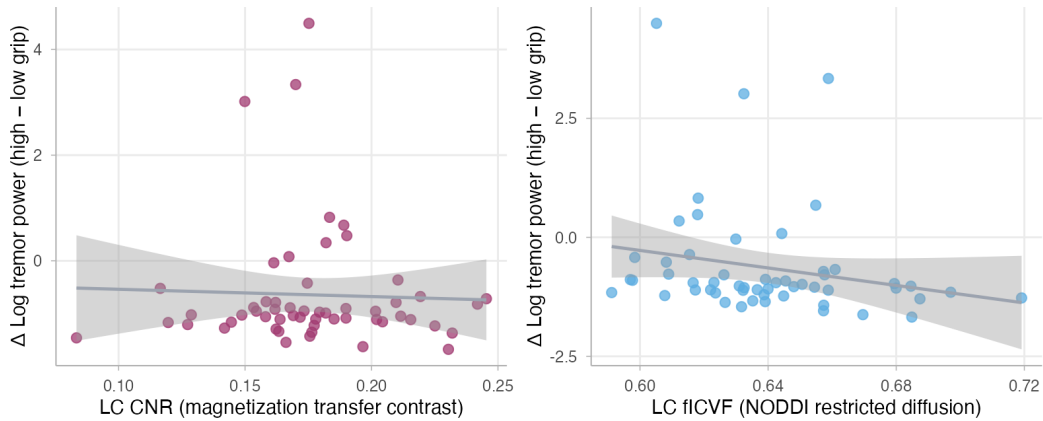


Figure 10: Exploratory LC fICVF associations with load-related tremor reactivity and perceptual load cost.

Table 3: Complete fixed-effects tables for baseline and DSP-augmented models (high-grip trials; post-outlier-screen, $|\text{tremor } z| \geq 3.5$).

Model	Predictor	OR (95% CI)	p
Baseline (high grip)	Intercept	2.01 (1.62, 2.48)	< .001
Baseline (high grip)	Modality (visual vs. auditory)	1.69 (1.47, 1.95)	< .001
Baseline (high grip)	Stimulus difficulty (linear)	4.91 (3.97, 6.07)	< .001
Baseline (high grip)	Stimulus difficulty (quadratic)	30.80 (25.29, 37.50)	< .001
Baseline (high grip)	Stimulus difficulty (cubic)	0.16 (0.14, 0.19)	< .001
Baseline (high grip)	Stimulus difficulty (4th order)	1.75 (1.53, 2.00)	< .001
DSP-augmented (high grip)	Intercept	1.97 (1.59, 2.44)	< .001
DSP-augmented (high grip)	Modality (visual vs. auditory)	1.70 (1.47, 1.96)	< .001
DSP-augmented (high grip)	Stimulus difficulty (linear)	4.92 (3.98, 6.08)	< .001
DSP-augmented (high grip)	Stimulus difficulty (quadratic)	30.87 (25.34, 37.59)	< .001
DSP-augmented (high grip)	Stimulus difficulty (cubic)	0.16 (0.14, 0.19)	< .001
DSP-augmented (high grip)	Stimulus difficulty (4th order)	1.75 (1.53, 1.99)	< .001
DSP-augmented (high grip)	Tracking power 0.5–3 Hz (z)	1.03 (0.95, 1.13)	0.452
DSP-augmented (high grip)	Tremor power 8–12 Hz (z)	0.64 (0.46, 0.89)	0.008

5 Spectral Outlier Robustness

The primary analysis excluded trials whose tremor-band power z-score (computed within each grip arm) exceeded ± 3.5 SD. Accuracy labels were not used in defining this screen. The table below shows results under four alternative screening rules to assess sensitivity of the main finding.

Table 4: Sensitivity of the high-grip DSP model to alternative spectral outlier screens. Tremor OR and 95% CI from the DSP-augmented mixed-effects logistic model; LRT p compares the DSP-augmented vs. baseline model. The primary analysis ($|z| > 3.5$) is shown in bold.

Outlier rule	N re- moved	N trials	Tremor OR	95% CI	Tremor p	LRT p
None (no screen)	0	6911	0.96	[0.88, 1.04]	0.282	0.447
Exclude $ z > 4.0$	20	6891	0.66	[0.48, 0.89]	0.007	0.023
Exclude $z > 3.5$ (primary) (<i>primary</i>)	23	6888	0.64	[0.46, 0.89]	0.008	0.025
Exclude $ z > 3.0$	26	6885	0.65	[0.47, 0.91]	0.012	0.040
Winsorize ± 3.5	0	6911	0.69	[0.54, 0.89]	0.004	0.013

Interpretation: The negative tremor-accuracy association is statistically significant across all four screening approaches, including no-removal (winsorization) and more or less aggressive exclusion thresholds. The no-screen model (OR = 0.96, LRT $p = .447$) shows that the 23 removed high-grip trials exerted high leverage: their extreme x-values (max $z = 48.5$) with moderate accuracy pulled the slope toward zero. Their removal is justified on physiological grounds — no isometric grip-force epoch should produce tremor-band power 48 SD above the population mean — independent of accuracy outcomes.

Outlier distribution: The 23 removed high-grip trials came from 4 participants (BAP139, BAP166, BAP171, BAP176; IDs shown here for reproducibility), distributed across both auditory ($n = 12$) and visual ($n = 11$) modalities, suggesting scattered recording artifacts rather than a single systematic run failure.

6 Reproducibility

Analysis notebooks live in `analysis/`:

Notebook	Purpose
01_ingest.qmd	Trial table construction and spectral feature extraction
02_cognitive_motor_model.qmd	Mixed-effects models and LC brain-behavior analyses

Notebook	Purpose
03_hero_visuals.qmd	Figure generation for manuscript and portfolio

Render from the repository root:

```
quarto render analysis/01_ingest.qmd
quarto render analysis/02_cognitive_motor_model.qmd
quarto render analysis/03_hero_visuals.qmd
quarto render manuscript/manuscript.qmd
quarto render manuscript/manuscript-biorxiv.qmd --to biorxiv-typst
quarto render manuscript/supplementary.qmd
```

Local data paths are configured in `config/paths.local.yaml` (not tracked in git).